Knowledge Base Article Collection

# Connectivity & Integration

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| **Document Type** | Knowledge Base Article Collection |
| **Product** | EchoCRS — Integration Gateway |
| **Audience** | L1 / L2 Support Engineers (internal) |
| **Articles** | KB-001 through KB-008 |
| **Topics Covered** | Network connectivity, SSL/TLS, overbooking, code mapping |
| **Version** | 1.0 |

## About This Collection

This collection contains eight knowledge base articles covering the most common connectivity and integration issues encountered in the EchoCRS Integration Gateway layer. The Integration Gateway is the unified connection point between EchoCRS and a property's Property Management System (PMS), handling all inbound and outbound reservation, inventory, and rate traffic.

Articles are organized by problem category and follow a consistent dual-audience format: an Internal Resolution section for L1/L2 support engineers, followed by a Customer-Facing View block containing approved language for client communication.

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| ℹ NOTE | Each article includes both an Internal Resolution section (for support engineers) and a Customer-Facing View section (approved language for client communication). These sections serve different audiences and must not be mixed. Do not share internal resolution steps, escalation paths, or VPN-gated resources with clients. |

## Article Index

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| KB-002 | Integration Gateway — PMS Vendor/Property Network Failure | Connectivity | Medium |
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| KB-006 | Integration Gateway — SSL Certificate Issue | SSL / TLS | Medium |

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| **KB-007** | Integration Gateway — Group PMS Code Mapping | Configuration | Medium |
| **KB-008** | Integration Gateway — Inventory & Rate PMS Code Mapping | Configuration | Medium |

## Category: Network Connectivity Failures

The following articles cover scenarios where connectivity between EchoCRS and a property's PMS is disrupted through the Integration Gateway. Failures may originate internally (within EchoCRS infrastructure) or externally (at the property, PMS vendor, or network level). Initial triage determines failure origin before resolution steps are applied.

KB-001

### EchoCRS Integration Gateway — Internal Network Failure

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|---|---|
| **Category** | Connectivity — Internal Origin |
| **Audience** | L1 / L2 Support Engineers |
| **Priority** | 2 — High |
| **Keywords** | integration gateway, network, failure, internal, system status |
| **Tools Referenced** | System Status Dashboard, Kibana |

#### Problem Description

Client reports a connectivity failure when submitting updates via their PMS. EchoCRS returns an error per the PMS vendor. After initial review, the failure appears to originate within the EchoCRS Integration Gateway rather than at the property or vendor side.

#### Internal Resolution

1. Check the internal system status dashboard to confirm whether a known outage or degradation is currently affecting the Integration Gateway.
2. If an internal issue is confirmed:
   - → Do not attempt further client-side triage — this is not a property or vendor issue.
   - → Escalate immediately to L3 for resolution.
   - → Notify the client that the issue has been identified internally and is under investigation. Do not share internal system details or status dashboard information.
3. If no internal issue is confirmed, re-evaluate the failure origin.
   - → If logs in Kibana show errors returning from the property or vendor endpoint, refer to KB-002 for external connectivity investigation.
   - → If the origin remains unclear, engage L2 before proceeding.

**Customer-Facing View**

*Approved language for client communication:*

- We have identified a connectivity issue on our end affecting your integration and our team is actively investigating.
- No action is required from you or your PMS vendor at this time.
- We will follow up with a status update as soon as the issue is resolved or we have additional information to share.

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KB-002

## EchoCRS Integration Gateway — PMS Vendor / Property Network Failure

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|---|---|
| **Category** | Connectivity — External Origin |
| **Audience** | L1 / L2 Support Engineers |
| **Priority** | 2 — High |
| **Keywords** | integration gateway, network, failure, firewall, property, PMS vendor |
| **Tools Referenced** | Kibana, Connectivity Test Tool (VPN required) |

### Problem Description

EchoCRS returns a connectivity error when sending data through the Integration Gateway. Internal EchoCRS systems are confirmed healthy. The failure appears to originate at the property's network, the PMS vendor's infrastructure, or the connection point between them.

### Internal Resolution

1. Review reservation and integration logs in Kibana to confirm an error is being returned and to identify the error type and origin point.
2. Using the Connectivity Test Tool (VPN connection required), run a connectivity test against the property's endpoint to confirm the failure is external.
3. Once the failure is confirmed as external, advise the property to verify the following:
   - → Internet connection is stable and active.
   - → Firewall rules allow inbound and outbound traffic to and from the EchoCRS integration servers.
   - → Provide the required IP addresses and port numbers and request confirmation that firewall settings are configured correctly on their side.
4. If the property confirms their configuration is correct, engage the PMS vendor to:
   - → Check for any active outages or network blocks on their side.
   - → Confirm no recent changes were made to their network or server configuration.
   - → Verify their endpoint is reachable and responding within expected timeframes.
5. If both the property and PMS vendor confirm no issues on their end and the failure persists, escalate to L3 with Kibana log exports and connectivity test results attached.

**Customer-Facing View**

*Approved language for client communication:*

- Our systems are operational, but we're detecting a connectivity issue between our Integration Gateway and your property's network.
- Please verify that your internet connection is stable and that your firewall allows traffic to and from our integration servers. We can provide the required IP addresses and port numbers.
- If your network configuration appears correct, we recommend contacting your PMS vendor to check for outages or recent changes on their side.
- Please let us know what you find and we will continue to assist from our end.

## Category: Reservation Overbooking

The following article covers overbooking scenarios reported through the Integration Gateway. This article provides a quick-reference resolution path. For full investigation workflow including group booking logic, escalation decision matrix, and timestamp analysis guidance, refer to the EchoCRS Reservation Overbooking Troubleshooting Guide.

KB-003

### EchoCRS Integration Gateway — Reservation Overbooking

| | |
|---|---|
| **Category** | Inventory — Overbooking |
| **Audience** | L1 / L2 Support Engineers |
| **Priority** | 3 — Medium |
| **Keywords** | integration gateway, reservation, overbooking, inventory, ARI, OTA |
| **Tools Referenced** | Kibana, Reservation Reports, Channel Management Team |

### Problem Description

Client reports that a reservation was accepted despite no available inventory, resulting in an overbooking. The booking was processed through the Integration Gateway. Investigation should determine whether the overbooking is valid based on inventory state at the time of booking.

### Internal Resolution — Path A: OTA Timing Window

*Use this path when the booking source is an OTA (Online Travel Agency).*

1. In Kibana, locate the last inventory message received prior to the booking. Record the exact timestamp.
2. If the last pre-booking message shows availability, verify the booking source.
   - → If the booking source is an OTA, proceed to Step 3.
   - → If the last pre-booking message shows 0 availability prior to the booking, proceed to Path B.
3. Check the time elapsed between the last inventory message and the booking timestamp.

→ OTA platforms can take up to 15 minutes to receive and process ARI (Availability, Rates, and Inventory) updates.

→ If the booking was processed within this 15-minute window, the overbooking may be attributable to ARI propagation delay — not a system error.

→ For bookings via Channel Management: engage the channel management team to review OTA submission logs.

→ For Expedia Quick Connect bookings: submission-level access is not available. Document findings and escalate.

### Internal Resolution — Path B: Cancellation or Configuration

*Use this path when no OTA timing explanation applies, or after Path A is ruled out.*

**1.** In Kibana, capture the last inventory message prior to the booking and the first inventory message after the booking. Record both timestamps.

**2.** If the last pre-booking message shows 0 availability, proceed to Step 3. If it shows availability, advise the client based on the inventory data and document findings.

**3.** Check for cancellations that occurred between the last pre-booking inventory message timestamp and the booking timestamp.

→ If cancellations are found: the overbooking is not valid. Advise the client with the supporting cancellation and inventory data.

→ If no cancellations are found, proceed to Step 4.

**4.** Check the *Include Hold Bookings in Available Count* property setting.

→ If the setting is NOT enabled: enable it, advise the client that a configuration change has been applied, and note that this should prevent recurrence.

→ If the setting IS already enabled: this case is a candidate for L3 escalation. Compile Kibana log exports, timestamp comparison, and configuration screenshot before escalating.

| ℹ NOTE | For full step-by-step investigation including group booking scenarios and the escalation decision matrix, refer to the EchoCRS Reservation Overbooking Troubleshooting Guide. |
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**Customer-Facing View**

*Approved language for client communication:*

- We've received your report and are reviewing the reservation and inventory logs through our Integration Gateway.
- We will confirm whether the booking reflects a valid inventory state at the time it was processed.
- If a configuration adjustment is needed on our end, we will apply it and notify you. If further investigation is required, we will follow up with our findings and next steps.

## Category: SSL / TLS & URL Alerts

The following articles address endpoint verification failures and SSL/TLS certificate issues that interrupt the Integration Gateway's connection to the PMS vendor's server. These errors are typically vendor-side but require internal confirmation before the client is engaged.

KB-004

### EchoCRS Integration Gateway — URL Alert: Connection Timed Out

| | |
|---|---|
| **Category** | Connectivity — Timeout |
| **Audience** | L1 / L2 Support Engineers |
| **Priority** | 3 — Medium |
| **Keywords** | integration gateway, timeout, URL alert, network, server, unresponsive |
| **Tools Referenced** | Connectivity Test Tool (VPN required) |

#### Problem Description

The Integration Gateway generates a URL alert indicating a connection timeout when attempting to reach the PMS vendor's server. This typically indicates a slow or unresponsive PMS server, but may also be caused by firewall rules, high network traffic, or server resource issues on the vendor's side.

#### Internal Resolution

1. Ask the property and PMS vendor to confirm their systems are currently online and responsive.
2. Request that the PMS vendor verify the following on their side:
   - → No active outages affecting their server or network.
   - → Firewall rules are not blocking inbound requests from the EchoCRS Integration Gateway.
   - → Server resources are not under excessive load (CPU, memory, or connection limits).
   - → Network traffic volume is within normal range.
3. If all of the above appear normal, request a server restart from the PMS vendor.
4. If the timeout persists after a restart, escalate to L2 with the timeout alert logs and vendor response documentation attached.

> **Customer-Facing View**
>
> *Approved language for client communication:*
>
> › We're seeing a connection timeout when our Integration Gateway attempts to reach your PMS vendor's server.
>
> › Please contact your PMS vendor and ask them to confirm their server is online and responsive, check for any active outages, and verify their firewall is not blocking traffic from our integration.
>
> › If everything appears normal on their end, a server restart may resolve the issue.
>
> › Please let us know the outcome and we will continue to monitor from our side.

KB-005

# EchoCRS Integration Gateway — URL Alert: Cannot Verify

| | |
|---|---|
| **Category** | SSL / TLS — Certificate Verification |
| **Audience** | L1 / L2 Support Engineers |
| **Priority** | 3 — Medium |
| **Keywords** | integration gateway, SSL, TLS, certificate, cannot verify, URL alert, whitelist |
| **Tools Referenced** | Connectivity Test Tool (VPN required), Endpoint Whitelist (VPN required) |

## Problem Description

The Integration Gateway generates a URL alert indicating it cannot verify the PMS vendor's endpoint. This typically points to an SSL/TLS certificate issue, an expired certificate, or an endpoint URL that is not on the approved whitelist.

## Internal Resolution

1. Confirm the correct endpoint URL is configured in EchoCRS for the property.
2. Verify the endpoint URL appears on the approved endpoint whitelist (VPN connection required to access).
3. Using the Connectivity Test Tool (VPN required), run a certificate validation test against the endpoint to determine whether the SSL certificate is valid and not expired.
4. If the certificate is invalid or expired:
   - → Advise the client to contact their PMS vendor immediately.
   - → The PMS vendor must update or reissue their SSL certificate before the connection can be re-established.
5. If the endpoint URL is not on the approved whitelist, work with internal teams to verify whether the URL is valid and should be added.
6. If the issue remains unresolved after completing Steps 1–5, escalate to L3 with connectivity test results and endpoint verification findings attached.

**⚠ WARNING**

Do not share the whitelist URL, internal VPN-gated resources, or internal endpoint verification results with clients or PMS vendors. All internal verification steps must be completed by support staff only.

### Customer-Facing View

*Approved language for client communication:*

- Our Integration Gateway is unable to verify the connection to your PMS vendor's server. This is typically caused by an SSL/TLS certificate issue on the vendor's side.
- Please contact your PMS vendor and ask them to check whether their SSL certificate is current and valid. If the certificate has expired or is incorrectly configured, they will need to reissue it.

- Please let us know once they've reviewed their certificate and we will re-test the connection from our side.

KB-006

## EchoCRS Integration Gateway — SSL Certificate Issue

| | |
|---|---|
| **Category** | SSL / TLS — Certificate |
| **Audience** | L1 / L2 Support Engineers |
| **Priority** | 3 — Medium |
| **Keywords** | integration gateway, SSL, TLS, certificate, expired, invalid |
| **Tools Referenced** | Connectivity Test Tool (VPN required), Endpoint Whitelist (VPN required) |

### Problem Description

The Integration Gateway reports an SSL certificate error when connecting to the PMS vendor endpoint. This is typically caused by an expired, invalid, or incorrectly configured certificate on the PMS vendor's server.

### Internal Resolution

Follow the same resolution steps as KB-005. This scenario presents with the same root causes and requires the same endpoint verification, whitelist confirmation, and certificate validation workflow.

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| ℹ NOTE | If KB-005 (Cannot Verify) and KB-006 (SSL Certificate Issue) symptoms present simultaneously for the same property, treat as a single case. One connectivity test covering both the alert and the certificate error is sufficient. Document both symptoms in the case notes. |

**Customer-Facing View**

*Approved language for client communication:*

- We've detected an SSL certificate error on the connection between our Integration Gateway and your PMS vendor's server.
- Please contact your PMS vendor and request that they review and reissue their SSL certificate.
- Once they've confirmed the certificate has been updated, please let us know and we'll re-test the connection from our side.

# Category: PMS Code Mapping

The following articles address configuration mismatches between EchoCRS and the PMS vendor system. Code mapping issues occur when room type codes, rate codes, or group codes in EchoCRS do not match the corresponding values configured in the PMS. These mismatches prevent group, inventory, rate, and reservation data from transmitting correctly through the Integration Gateway.

| ℹ NOTE | All code matching in EchoCRS is case-sensitive. A code entered as 'KING' will not match 'King' or 'king'. Always confirm an exact character-for-character match — including capitalization — when advising clients on code corrections. |
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KB-007

## EchoCRS Integration Gateway — Group PMS Code Mapping

| | |
|---|---|
| **Category** | Configuration — Code Mapping |
| **Audience** | L1 / L2 Support Engineers |
| **Priority** | 3 — Medium |
| **Keywords** | integration gateway, group code, mapping, configuration, case-sensitive, PMS |
| **Tools Referenced** | EchoCRS Admin Console, PMS Configuration |

### Problem Description

Group reservations are failing or not transmitting correctly through the Integration Gateway to the PMS. Investigation reveals a mismatch between group block codes configured in EchoCRS and the corresponding codes in the PMS system.

### Internal Resolution

1. In EchoCRS, locate the group block codes associated with the affected property.
2. Compare each group code against the values configured in the PMS (based on log review). Confirm exact character-for-character match, including case.
3. If a mismatch is found, resolve using one of the following approaches:
   - → Update the group code mapping in EchoCRS to match the PMS, or
   - → Advise the client to contact their PMS vendor to confirm the correct code values and update them in the PMS accordingly.
4. After the correction is made, request a test update from the client to confirm the mapping is functioning correctly end-to-end.

**Customer-Facing View**

*Approved language for client communication:*

- We've identified a mismatch between the group codes configured in our system and those in your PMS, which is preventing group reservations from transmitting correctly.
- Please contact your PMS vendor to confirm the exact group codes currently configured on their side, then share those values with us so we can update the mapping.
- Alternatively, if you have direct access to update the codes in your PMS, please ensure they match exactly what is configured in EchoCRS — these comparisons are case-sensitive.
- Once the correction is made, please submit a test update so we can confirm the issue is resolved.

KB-008

# EchoCRS Integration Gateway — Inventory & Rate PMS Code Mapping

| | |
|---|---|
| **Category** | Configuration — Code Mapping |
| **Audience** | L1 / L2 Support Engineers |
| **Priority** | 3 — Medium |
| **Keywords** | integration gateway, inventory, rate code, room type, mapping, configuration, case-sensitive |
| **Tools Referenced** | EchoCRS Admin Console, PMS Configuration |

## Problem Description

Inventory updates or rate changes submitted from the PMS are not being applied correctly in EchoCRS through the Integration Gateway. Investigation reveals a code mismatch between room type or rate codes in EchoCRS and the corresponding values in the PMS.

## Internal Resolution — Inventory Code Mapping

*Note: Inventory does not require a separate PMS code mapping table in EchoCRS. Room type codes are matched directly between the two systems.*

1. Confirm all room types exist in EchoCRS, are active and not archived or disabled.
2. Compare the PMS room codes assigned to each room type in EchoCRS against the codes in the PMS system (based on log review). Confirm exact match including case.
3. If a mismatch is found:
   - → Update the PMS room code in EchoCRS to match the PMS, or
   - → Advise the client to contact their PMS vendor to confirm the correct values and update accordingly.

## Internal Resolution — Rate Code Mapping

1. In EchoCRS, locate the rate plan associated with the affected rate.
2. Confirm the PMS rate code assigned to the rate plan matches the corresponding code in the PMS mapping table exactly, including case.
3. If a mismatch or missing mapping is found:
   - → The client must update the mapping table in their PMS to reflect the correct rate code, or
   - → Request the correct PMS rate code from their PMS vendor and update the EchoCRS configuration accordingly.
4. After any correction is made, request a test ARI update to confirm rates and inventory are transmitting correctly through the Integration Gateway.

### Customer-Facing View

*Approved language for client communication:*

- We've identified a code mismatch between our system and your PMS that is preventing inventory or rate updates from applying correctly through the Integration Gateway.
- Please verify that the room type codes and rate codes in your PMS match exactly what is configured in EchoCRS — these comparisons are case-sensitive, so capitalization must match.
- If you're unsure of the correct codes on your PMS side, please contact your PMS vendor to confirm the current values.
- Once the codes are corrected, please submit a test update so we can confirm everything is transmitting correctly.

*This document is intended for internal support team use only. Customer-facing language in each article has been reviewed for clarity and is approved for direct client communication. Do not share internal resolution steps, VPN-gated resources, escalation criteria, or system status information with clients or third parties. Review annually or following significant product or integration changes.*